

MH3100 Real Analysis I

Tutorial 1 Solutions

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Overview

- **Lecture coverage:** Lectures 1–3, namely the construction of $\mathbb{R}$, the Axiom of Completeness, suprema, infima, maxima, minima, the Nested Interval Property, and the Archimedean Property.
- **Questions:** 8 questions in total, namely Q1 through Q5 and Additional Questions A1 through A3, extending the official answers and hints for Tutorial 1.
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$, and 0 is *not* a natural number in this course; sup, inf, max, min are written with the built-in operators; ε denotes ε .

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Question 1 (Suprema, infima, maxima, minima of four sets)

Problem

Compute, without proofs, the suprema and infima (if exist) of the following sets, and determine if the suprema and infima (if exist) are, respectively, maxima and minima of the respective sets.

(a) $A = \left\{ \frac{m}{n} : m, n \in \mathbb{N} \text{ with } m \leq n \right\}.$

(b) $B = \left\{ \frac{(-1)^m}{n} : m, n \in \mathbb{N} \right\}.$

(c) $C = \left\{ \frac{n}{3n+1} : n \in \mathbb{N} \right\}.$

(d) $D = \left\{ \frac{m}{m+n} : m, n \in \mathbb{N} \right\}.$

(Tutorial 1, Q1.)

Solution

Official hint. The question explicitly asks for answers “without proofs”: $\sup A = 1$ is a maximum, $\inf A = 0$ is not a minimum; $\sup B = 1$ is a maximum, $\inf B = -1$ is a minimum; $\sup C = \frac{1}{3}$ is not a maximum, $\inf C = \frac{1}{4}$ is a minimum; $\sup D = 1$ is not a maximum, $\inf D = 0$ is not a minimum.

Beyond the hints. An exam-standard treatment still needs a full justification of each claim: an upper/lower-bound verification, and either an explicit attaining element (for the max/min cases) or an ε -Criterion / Archimedean Property argument that the bound is approached but never attained. We supply this below, part by part.

Part (a): $A = \left\{ \frac{m}{n} : m, n \in \mathbb{N}, m \leq n \right\}$

Idea. (Scratch work – not part of the written answer.) Since $m \leq n$ with $m, n \in \mathbb{N}$, every element of A is at most 1, and $m = n$ (e.g. $m = n = 1$) attains 1. For the infimum, fixing $m = 1$ and letting $n \rightarrow \infty$ pushes m/n down towards 0, but $m/n > 0$ always since $m \geq 1$.

Proof (supremum). Let $a = m/n \in A$ with $m, n \in \mathbb{N}$ and $m \leq n$. Since $n > 0$, dividing $m \leq n$ by n gives $a = m/n \leq 1$. Therefore 1 is an upper bound for A . Moreover, taking $m = n = 1$ gives the element $1/1 = 1 \in A$. Since 1 is both an upper bound for A and an element of A , 1 is the maximum of A by the definition of maximum, and hence also the supremum of A by the Remark relating maxima to suprema.

$\sup A = \max A = 1.$

□

Proof (infimum). Every element $m/n \in A$ satisfies $m \geq 1$ and $n \geq 1$, so $m/n > 0$; hence 0 is a lower bound for A . We show 0 is the greatest lower bound using the ε -Criterion for Suprema and Infima (infimum version): let $\varepsilon > 0$ be given. By the Archimedean Property there exists $n \in \mathbb{N}$ with $1/n < \varepsilon$. Taking $m = 1 \leq n$ gives the element $1/n \in A$ with $1/n < 0 + \varepsilon$. Since $\varepsilon > 0$ was arbitrary, no lower bound greater than 0 exists, so 0 is the infimum of A . Finally, $0 \notin A$, since every element of A has the form m/n with $m \geq 1, n \geq 1$, so $m/n \geq 1/n > 0$. Therefore 0 is not attained, so it is not the minimum of A .

$$\inf A = 0, \quad 0 \text{ is not the minimum of } A.$$

□

Part (b): $B = \left\{ \frac{(-1)^m}{n} : m, n \in \mathbb{N} \right\}$

Idea. (*Scratch work – not part of the written answer.*) Splitting on the parity of m : for even m , $(-1)^m = 1$, so those elements are $1/n$; for odd m , $(-1)^m = -1$, so those elements are $-1/n$. Hence $B = \{1/n : n \in \mathbb{N}\} \cup \{-1/n : n \in \mathbb{N}\}$, and the extreme values 1 and -1 occur at $n = 1$.

Proof (supremum). Let $b = (-1)^m/n \in B$. Since $n \geq 1$, $1/n \leq 1$, so $b \leq |b| = 1/n \leq 1$; hence 1 is an upper bound for B . Taking $m = 2$ (so $(-1)^m = 1$) and $n = 1$ gives the element $(-1)^2/1 = 1 \in B$. Since 1 is an upper bound for B and $1 \in B$, 1 is the maximum of B by definition, and therefore also the supremum of B .

$$\sup B = \max B = 1.$$

□

Proof (infimum). By the same reasoning, $b = (-1)^m/n \geq -1/n \geq -1$ for all $m, n \in \mathbb{N}$, so -1 is a lower bound for B . Taking $m = 1$ (so $(-1)^m = -1$) and $n = 1$ gives the element $-1 \in B$. Since -1 is a lower bound for B and $-1 \in B$, -1 is the minimum of B , and therefore also the infimum of B .

$$\inf B = \min B = -1.$$

□

Part (c): $C = \left\{ \frac{n}{3n+1} : n \in \mathbb{N} \right\}$

Idea. (*Scratch work – not part of the written answer.*) Write $c_n = n/(3n+1)$. Comparing consecutive terms shows $\{c_n\}$ is strictly increasing, so the infimum should be the first term $c_1 = 1/4$ (attained), and the supremum should be the limiting value $1/3$ (not attained, since $c_n < 1/3$ for every finite n).

Proof (infimum). We first show $\{c_n\}$ is strictly increasing, i.e. $c_{n+1} > c_n$ for every $n \in \mathbb{N}$. Compute

$$\begin{aligned} c_{n+1} - c_n &= \frac{n+1}{3n+4} - \frac{n}{3n+1} \\ &= \frac{(n+1)(3n+1) - n(3n+4)}{(3n+4)(3n+1)} \\ &= \frac{3n^2 + 4n + 1 - 3n^2 - 4n}{(3n+4)(3n+1)} \\ &= \frac{1}{(3n+4)(3n+1)}. \end{aligned}$$

Since $n \geq 1$, both factors in the denominator are positive, so $c_{n+1} - c_n > 0$, i.e. $c_{n+1} > c_n$. By induction, $c_n \geq c_1$ for every $n \in \mathbb{N}$, with equality exactly when $n = 1$. Hence $c_1 = 1/4$ is a lower bound for C , and since $c_1 \in C$, $1/4$ is the minimum of C , and therefore also the infimum of C .

$$\boxed{\inf C = \min C = \frac{1}{4}.}$$

□

Proof (supremum). For every $n \in \mathbb{N}$, $3n < 3n+1$, so dividing by the positive number $3(3n+1)$ gives $\frac{n}{3n+1} = \frac{3n}{3(3n+1)} < \frac{3n+1}{3(3n+1)} = \frac{1}{3}$. Hence $1/3$ is an upper bound for C , and no element of C equals $1/3$. We show $1/3$ is the least upper bound via the ε -Criterion for Suprema and Infima: let $\varepsilon > 0$ be given. By the Archimedean Property there exists $n \in \mathbb{N}$ with $1/n < \varepsilon$. Then

$$\begin{aligned} \frac{1}{3} - \frac{n}{3n+1} &= \frac{(3n+1) - 3n}{3(3n+1)} \\ &= \frac{1}{3(3n+1)} \\ &< \frac{1}{9n} \\ &= \frac{1}{9} \cdot \frac{1}{n} \\ &< \frac{1}{9} \varepsilon < \varepsilon, \end{aligned}$$

using $3(3n+1) > 9n$ in the third line and $1/n < \varepsilon$ in the fifth, so the element $c_n = n/(3n+1) \in C$ satisfies $1/3 - \varepsilon < c_n$. Since $\varepsilon > 0$ was arbitrary, $1/3$ is the supremum of C ; and since no element of C equals $1/3$, it is not the maximum.

$$\boxed{\sup C = \frac{1}{3}, \quad \frac{1}{3} \text{ is not the maximum of } C.}$$

□

Part (d): $D = \left\{ \frac{m}{m+n} : m, n \in \mathbb{N} \right\}$

Idea. (*Scratch work – not part of the written answer.*) Fixing $n = 1$ and letting $m \rightarrow \infty$ pushes $m/(m+n) \rightarrow 1$ without ever reaching it (since $n \geq 1$ always leaves a positive gap). Symmetrically, fixing $m = 1$ and letting $n \rightarrow \infty$ pushes $m/(m+n) \rightarrow 0$ without reaching it.

Proof (supremum). Let $d = m/(m+n) \in D$ with $m, n \in \mathbb{N}$. Since $n \geq 1$, $m+n > m$, so dividing by the positive number $m+n$ gives $d = m/(m+n) < (m+n)/(m+n) = 1$. Hence 1 is an upper bound for D , and no element of D equals 1. We show 1 is the least upper bound via the ε -Criterion for Suprema and Infima: let $\varepsilon > 0$ be given. By the Archimedean Property there exists $m \in \mathbb{N}$ with $1/m < \varepsilon$. Taking $n = 1$, the element $d = m/(m+1) \in D$ satisfies

$$1 - d = 1 - \frac{m}{m+1} = \frac{1}{m+1} < \frac{1}{m} < \varepsilon,$$

so $1 - \varepsilon < d$. Since $\varepsilon > 0$ was arbitrary, 1 is the supremum of D ; since no element attains it, it is not the maximum.

$$\boxed{\sup D = 1, \quad 1 \text{ is not the maximum of } D.}$$

□

Proof (infimum). By a symmetric argument, $d = m/(m+n) > 0$ for all $m, n \in \mathbb{N}$ (since $m \geq 1$), so 0 is a lower bound for D , and no element of D equals 0. We show 0 is the greatest lower bound via the ε -Criterion for Suprema and Infima: let $\varepsilon > 0$ be given. By the Archimedean Property there exists $n \in \mathbb{N}$ with $1/n < \varepsilon$. Taking $m = 1$, the element $d = 1/(1+n) \in D$ satisfies $d = 1/(1+n) < 1/n < \varepsilon$, so $d < 0 + \varepsilon$. Since $\varepsilon > 0$ was arbitrary, 0 is the infimum of D ; since no element attains it, it is not the minimum.

$$\boxed{\inf D = 0, \quad 0 \text{ is not the minimum of } D.}$$

□

Common slip. A frequent error is to claim $\sup A$ or $\sup D$ is a maximum merely because the set “looks like” it approaches 1 closely; the definition of maximum requires the bound to be an *element* of the set, so an explicit witnessing element (attained case) or an ε -argument (unattained case) is mandatory, not optional. Another common slip is choosing n depending on the wrong quantity – the Archimedean Property produces n from ε (or from a), never the reverse.

Question 2 (Supremum of a union: $\max\{\sup A, \sup B\} = \sup(A \cup B)$)

Problem

Prove that if the sets $A, B \subseteq \mathbb{R}$ are nonempty and bounded above, then

$$\max\{\sup A, \sup B\} = \sup(A \cup B).$$

(Tutorial 1, Q2; this is also the sample midterm Q3.)

Solution

Official hint. Prove two inequalities using only the definition of supremum: (a) $\sup(A \cup B)$ is an upper bound for A and for B , so $\sup A \leq \sup(A \cup B)$ and $\sup B \leq \sup(A \cup B)$; (b) $\max\{\sup A, \sup B\}$ is an upper bound for $A \cup B$, so $\sup(A \cup B) \leq \max\{\sup A, \sup B\}$.

Beyond the hints. The hint does not spell out why $\sup(A \cup B)$ exists in the first place, nor the explicit chain of implications tying each part together; both are filled in below. Because Q2 recurs as Question 3 of the sample midterm (a “definition of ... and use the definition to show ...” question with a ban on named shortcut theorems), Method 1 below is written as a model exam answer that uses *only* the definition of supremum – no ε -Criterion.

Method 1: Two inequalities from the definition of supremum (model exam answer)

Idea. (*Scratch work – not part of the written answer.*) Existence first: $A \cup B$ is nonempty and bounded above (by $\max\{\sup A, \sup B\}$), so its supremum exists. Then chase the definition of supremum in both directions: “an upper bound for the bigger set restricts to an upper bound for each piece”, and “an upper bound for each piece assembles into an upper bound for the union”.

Proof. Since A and B are nonempty and bounded above, the Axiom of Completeness guarantees that $\sup A$ and $\sup B$ exist. Let $s = \sup A$ and $t = \sup B$, and let $M = \max\{s, t\}$.

First we show $A \cup B$ is nonempty and bounded above. Since A is nonempty, $A \cup B$ is nonempty. For every $x \in A \cup B$, either $x \in A$, in which case $x \leq s \leq M$ since s is an upper bound for A ; or $x \in B$, in which case $x \leq t \leq M$ since t is an upper bound for B . In either case $x \leq M$, so M is an upper bound for $A \cup B$. Therefore $A \cup B$ is nonempty and bounded above, so by the Axiom of Completeness, $\sup(A \cup B)$ exists.

Step 1: $\sup(A \cup B) \leq M$. We have just shown that $M = \max\{s, t\}$ is an upper bound for $A \cup B$. Since $\sup(A \cup B)$ is, by definition, not greater than any upper bound for $A \cup B$, it follows that $\sup(A \cup B) \leq M$.

Step 2: $M \leq \sup(A \cup B)$. Let $u = \sup(A \cup B)$. Since $A \subseteq A \cup B$, every $a \in A$ satisfies $a \in A \cup B$, hence $a \leq u$; thus u is an upper bound for A . Since $s = \sup A$ is not greater than any upper bound for A , we get $s \leq u$. By the identical argument with B in place of A (using $B \subseteq A \cup B$), we get $t \leq u$. Therefore $M = \max\{s, t\} \leq u = \sup(A \cup B)$.

Combining Steps 1 and 2, $\sup(A \cup B) \leq M$ and $M \leq \sup(A \cup B)$; therefore $\sup(A \cup B) = M = \max\{\sup A, \sup B\}$.

$$\boxed{\max\{\sup A, \sup B\} = \sup(A \cup B).}$$

□

Method 2: Via the ε -Criterion for Suprema and Infima

Idea. (*Scratch work – not part of the written answer.*) Method 1 already gives $\sup(A \cup B) \leq M$ directly from the definition. For the reverse inequality, it suffices to check that M satisfies the ε -Criterion for $A \cup B$: M is an upper bound (shown above), and for every $\varepsilon > 0$ some element of $A \cup B$ beats $M - \varepsilon$ – take it from whichever of A, B realises the larger supremum.

Proof. As in Method 1, $M = \max\{s, t\}$ is an upper bound for $A \cup B$, where $s = \sup A$, $t = \sup B$. We verify the second half of the ε -Criterion for Suprema and Infima. Let $\varepsilon > 0$ be given. Without loss of generality suppose $M = s$ (the case $M = t$ is symmetric, with B in place of A). Since $s = \sup A$, the ε -Criterion for A gives an element $a \in A$ with $s - \varepsilon < a$. Since $A \subseteq A \cup B$, this same a lies in $A \cup B$, so $M - \varepsilon = s - \varepsilon < a$ with $a \in A \cup B$. Since $\varepsilon > 0$ was arbitrary, M satisfies both halves of the ε -Criterion for $A \cup B$; therefore $M = \sup(A \cup B)$.

$$\boxed{\max\{\sup A, \sup B\} = \sup(A \cup B).}$$

□

Common slip. On the sample midterm, the question explicitly states “Do not use the ε -Criterion for suprema and infima”; Method 1 is therefore the required approach for full marks in that setting, not merely the safer one. A further slip is forgetting to justify that $A \cup B$ is nonempty and bounded above before invoking $\sup(A \cup B)$ – without this, the Axiom of Completeness cannot be invoked to assert the object even exists.

Question 3 ($\sup B = \inf A$ for the set of lower bounds)

Problem

Suppose that the set $A \subseteq \mathbb{R}$ is nonempty and bounded below, and define the set

$$B := \{b \in \mathbb{R} : b \text{ is a lower bound for } A\}.$$

Prove that $\sup B = \inf A$. (*Tutorial 1, Q3.*)

Solution

Official hint. (a) Show $\inf A \in B$, then deduce $\sup B \geq \inf A$. (b) Show $\sup B$ is no greater than every number in A , then deduce $\inf A \geq \sup B$.

Beyond the hints. The hint's step (a) presupposes that $\inf A$ already exists; the course defines the infimum “similarly” to the supremum in Lecture 2, but existence for a general nonempty set bounded below has not separately been proved by this point. We restructure the proof so that the existence of $\sup B$ (via the Axiom of Completeness, since B is nonempty and bounded above) is established first, and $\sup B$ is verified *directly* against the definition of infimum, which simultaneously proves that $\inf A$ exists and equals $\sup B$.

Method 1: Direct verification against the definition of infimum

Idea. (*Scratch work – not part of the written answer.*) B is nonempty (since A is bounded below) and bounded above (by any fixed element of A , since every lower bound of A is $\leq$ every element of A). So $\sup B$ exists. Now just check $\sup B$ satisfies both halves of the definition of infimum of A : it is a lower bound for A , and it is $\geq$ every other lower bound of A (the latter is automatic, since every lower bound of A lies in B by definition of B , and $\sup B$ is $\geq$ every element of B).

Proof. Since A is nonempty and bounded below, A has at least one lower bound, so B is nonempty. Fix any $a_0 \in A$. For every $b \in B$, b is a lower bound for A , so $b \leq a_0$ (because $a_0 \in A$). Hence a_0 is an upper bound for B , so B is nonempty and bounded above. By the Axiom of Completeness, $\sup B$ exists.

Step 1: $\sup B$ is a lower bound for A . Let $a \in A$ be arbitrary. For every $b \in B$, b is a lower bound for A and $a \in A$, so $b \leq a$. Thus a is an upper bound for B . Since $\sup B$ is not greater than any upper bound for B , $\sup B \leq a$. Since $a \in A$ was arbitrary, $\sup B \leq a$ for every $a \in A$; that is, $\sup B$ is a lower bound for A .

Step 2: $\sup B$ is the greatest lower bound for A . Let $c \in \mathbb{R}$ be any lower bound for A . By the definition of B , $c \in B$. Since $\sup B$ is an upper bound for B , $c \leq \sup B$. Since c was an arbitrary lower bound for A , every lower bound of A is $\leq \sup B$.

By Steps 1 and 2, $\sup B$ is a lower bound for A that is not less than any other lower bound for A ; this is exactly the definition of $\inf A$. Therefore $\inf A$ exists and

$$\boxed{\sup B = \inf A.}$$

□

Common slip. It is tempting to write “ $\inf A \in B$, so $\sup B \geq \inf A$ ” as the entire proof, as the official hint’s part (a) suggests; but this silently assumes $\inf A$ already exists, which for a general nonempty bounded-below set has not yet been established at this point in the course. Method 1 avoids the circularity by proving existence and equality simultaneously, directly from the Axiom of Completeness (which only guarantees suprema, applied here to B).

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Question 4 (Supremum of a sumset: $\sup(A+B) = \sup A + \sup B$)

Problem

Suppose that the sets $A, B \subseteq \mathbb{R}$ are nonempty and bounded above and that $s = \sup A$ and $t = \sup B$. Define the notation

$$A + B := \{a + b : a \in A \text{ and } b \in B\}.$$

- (a) Prove that $s + t$ is an upper bound for $A + B$.
- (b) Prove that if $u \in \mathbb{R}$ is an upper bound for $A + B$ and $a \in A$, then

$$t \leq u - a.$$

- (c) Hence deduce that $\sup(A + B) = s + t$.

(Tutorial 1, Q4.)

Solution

Official hint. (a) Write $c = a + b \in A + B$ and bound each summand. (b) Use the definition of upper bound to show $u - a$ is an upper bound for B . (c) Use the definition of supremum to show $s \leq u - t$ for every upper bound u of $A + B$.

Beyond the hints. The printed target of part (b) is $t \leq u - a$, while the hint's part (c) is phrased for the rearranged form $s \leq u - t$; neither the rearrangement nor the final assembly of (a) and (c) into “ $s + t$ is the *least* upper bound” is spelled out in the official hint. Both are supplied below.

Part (a)

Proof. Let $c \in A + B$ be arbitrary. By the definition of $A + B$, $c = a + b$ for some $a \in A$ and $b \in B$. Since $s = \sup A$ is an upper bound for A and $a \in A$, $a \leq s$. Since $t = \sup B$ is an upper bound for B and $b \in B$, $b \leq t$. Adding these two inequalities gives $c = a + b \leq s + t$. Since $c \in A + B$ was arbitrary, $s + t$ is an upper bound for $A + B$.

$s + t \text{ is an upper bound for } A + B.$

□

Part (b)

Idea. (*Scratch work – not part of the written answer.*) Fix $a \in A$. Every $b \in B$ pairs with this a to give $a + b \in A + B$, which is bounded above by u ; rearranging gives a bound on b that holds for every $b \in B$, i.e. an upper bound for B . Then invoke the least-upper-bound property of $t = \sup B$.

Proof. Let $u \in \mathbb{R}$ be an upper bound for $A + B$ and let $a \in A$. For every $b \in B$, the element $a + b$ lies in $A + B$ (by definition of $A + B$, since $a \in A$ and $b \in B$), so $a + b \leq u$, because u is an upper bound for $A + B$. Rearranging, $b \leq u - a$ for every $b \in B$; that is, $u - a$ is an upper bound for B . Since $t = \sup B$ is not greater than any upper bound for B , we conclude $t \leq u - a$.

$$t \leq u - a.$$

□

Part (c)

Idea. (*Scratch work – not part of the written answer.*) Part (a) already gives one half: $\sup(A + B) \leq s + t$ (once existence of $\sup(A + B)$ is secured). For the other half, take an arbitrary upper bound u of $A + B$, rearrange part (b)’s conclusion $t \leq u - a$ to $a \leq u - t$ for every $a \in A$ – so $u - t$ is an upper bound for A – and invoke the least-upper-bound property of $s = \sup A$ to get $s \leq u - t$, i.e. $s + t \leq u$. Since u was an arbitrary upper bound, $s + t$ is the *least* upper bound.

Proof. Since A and B are nonempty, $A + B$ is nonempty (pick any $a_0 \in A$, $b_0 \in B$; then $a_0 + b_0 \in A + B$). By part (a), $s + t$ is an upper bound for $A + B$, so $A + B$ is bounded above. By the Axiom of Completeness, $\sup(A + B)$ exists, and since $s + t$ is an upper bound, $\sup(A + B) \leq s + t$.

For the reverse inequality, let $u \in \mathbb{R}$ be an arbitrary upper bound for $A + B$. Let $a \in A$ be arbitrary. By part (b), $t \leq u - a$. Rearranging, $a \leq u - t$. Since $a \in A$ was arbitrary, $u - t$ is an upper bound for A . Since $s = \sup A$ is not greater than any upper bound for A , $s \leq u - t$, i.e. $s + t \leq u$. Since u was an arbitrary upper bound for $A + B$, $s + t$ is not greater than any upper bound for $A + B$.

Combining the two halves: $s + t$ is an upper bound for $A + B$ (part (a)) that is not greater than any other upper bound for $A + B$ (just shown); this is exactly the definition of supremum. Therefore

$$\sup(A + B) = s + t.$$

□

Common slip. Skipping the “ $A + B$ nonempty and bounded above” check before invoking $\sup(A + B)$ loses marks, since the Axiom of Completeness requires both hypotheses. A second slip is proving only $\sup(A + B) \leq s + t$ (i.e. only that $s + t$ is *an* upper bound) and stopping there – this shows $s + t$ is an upper bound, not that it is the *least* one, so the second half (using part (b)) is not optional.

Question 5 (An ε -characterisation of the supremum)

Problem

Prove that if $A \subseteq \mathbb{R}$ is nonempty and bounded above, and $s \in \mathbb{R}$ has the property that

$$\forall n \in \mathbb{N} \quad s + \frac{1}{n} \text{ is an upper bound for } A \text{ and } s - \frac{1}{n} \text{ is not an upper bound for } A,$$

then $s = \sup A$. (*Tutorial 1, Q5.*)

Solution

Official hint. (a) Split the hypothesis on s into two separate universal statements. (b) Assume the hypothesis on A and prove the contrapositive of the rest. (c) Use the Archimedean Property to prove the contrapositive statement.

Beyond the hints. This is the ε -Criterion for Suprema and Infima in a natural-number-indexed disguise; the hint does not name it or construct the contrapositive argument explicitly. We do so below, splitting into the two cases $s < \sup A$ and $s > \sup A$ that together negate $s = \sup A$.

Method 1: Contrapositive via the Archimedean Property

Idea. (*Scratch work – not part of the written answer.*) Since A is nonempty and bounded above, $s_0 := \sup A$ exists by the Axiom of Completeness; the goal becomes $s = s_0$. Suppose for contradiction $s \neq s_0$. If $s < s_0$: pick n with $1/n$ smaller than the gap $s_0 - s$, so $s + 1/n < s_0$; but $s + 1/n$ is supposed to be an upper bound for A , while s_0 is the *least* upper bound, so no upper bound can be less than s_0 – contradiction. If $s > s_0$: pick n with $1/n$ smaller than the gap $s - s_0$, so $s - 1/n > s_0$; since s_0 is an upper bound and $s - 1/n$ exceeds it, $s - 1/n$ is also an upper bound – but the hypothesis says it is not – contradiction.

Proof. Since A is nonempty and bounded above, the Axiom of Completeness guarantees that $s_0 := \sup A$ exists. We show $s = s_0$ by ruling out the two alternatives $s < s_0$ and $s > s_0$.

Case $s < s_0$. Let $\varepsilon = s_0 - s > 0$. By the Archimedean Property there exists $n \in \mathbb{N}$ with $1/n < \varepsilon = s_0 - s$. Rearranging, $s + 1/n < s_0$. By hypothesis, $s + 1/n$ is an upper bound for A . But $s_0 = \sup A$ is not greater than any upper bound for A , so every upper bound for A is at least s_0 . Since $s + 1/n$ is an upper bound for A with $s + 1/n < s_0$, this contradicts s_0 being no greater than every upper bound. Therefore the case $s < s_0$ is impossible.

Case $s > s_0$. Let $\varepsilon = s - s_0 > 0$. By the Archimedean Property there exists $n \in \mathbb{N}$ with $1/n < \varepsilon = s - s_0$. Rearranging, $s - 1/n > s_0$. Since s_0 is an upper bound for A and $s - 1/n > s_0$, every element $a \in A$ satisfies $a \leq s_0 < s - 1/n$, so $s - 1/n$ is also an upper bound for A . But by hypothesis, $s - 1/n$ is not an upper bound for A – a contradiction. Therefore the case $s > s_0$ is impossible.

Since both $s < s_0$ and $s > s_0$ are impossible, $s = s_0 = \sup A$.

$$\boxed{s = \sup A.}$$

□

Common slip. The index n in each case must be chosen *after* the gap $\varepsilon = |s - s_0|$ is fixed (the Archimedean Property produces n from a given positive number); choosing n first and then trying to define ε from it reverses the quantifier order and does not prove the claim. Another slip is treating the two hypotheses (“ $s + 1/n$ is an upper bound” and “ $s - 1/n$ is not”) as interchangeable; each is used in exactly one of the two cases above, not both in each case.

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Question A1 (Supremum of a countable union)

Problem

Suppose that $A_1, A_2, A_3, \dots$ is a list of nonempty sets, each of which is bounded above. Find a formula for $\sup(\bigcup_{n=1}^{\infty} A_n)$ in terms of $\sup A_1, \sup A_2, \sup A_3, \dots$ (*Tutorial 1, Additional Question A1.*)

Solution

Official hint. Answer only: $\sup(\bigcup_{n=1}^{\infty} A_n) = \sup\{\sup A_n : n \in \mathbb{N}\}$.

Beyond the hints. No proof strategy is given at all. We generalise the two-inequality argument of Q2 from a union of two sets to a countable union, and address a gap the underlying source material flags explicitly: the set $\{\sup A_n : n \in \mathbb{N}\}$ is not automatically bounded above, so the formula needs a case split.

Method 1: Two inequalities, with a case split on boundedness

Idea. (*Scratch work – not part of the written answer.*) Let $U = \bigcup_{n=1}^{\infty} A_n$ and let $S = \{\sup A_n : n \in \mathbb{N}\}$ (each $\sup A_n$ exists by the Axiom of Completeness, since each A_n is nonempty and bounded above). If S is bounded above, the Q2-style argument goes through verbatim with countably many sets in place of two. If S is unbounded above, then U is unbounded above too (an unbounded family of suprema forces elements of the A_n arbitrarily high), so both sides equal ∞ under the course's convention $\sup(\text{unbounded set}) = \infty$.

Proof. Let $U = \bigcup_{n=1}^{\infty} A_n$ and $S = \{\sup A_n : n \in \mathbb{N}\}$. Since each A_n is nonempty, U is nonempty; since each A_n is bounded above, $\sup A_n$ exists for every $n \in \mathbb{N}$ by the Axiom of Completeness, so S is a well-defined nonempty set of real numbers.

Case 1: S is bounded above. Let $M = \sup S$, which exists by the Axiom of Completeness. We show M is an upper bound for U : let $x \in U$ be arbitrary. Then $x \in A_n$ for some $n \in \mathbb{N}$, so $x \leq \sup A_n$ (since $\sup A_n$ is an upper bound for A_n), and $\sup A_n \leq M$ (since $\sup A_n \in S$ and $M = \sup S$ is an upper bound for S). Hence $x \leq M$. Since $x \in U$ was arbitrary, U is bounded above by M , so $\sup U$ exists.

We now show $\sup U = M$ via two inequalities. First, since M is an upper bound for U and $\sup U$ is not greater than any upper bound for U , $\sup U \leq M$. Second, for each $n \in \mathbb{N}$, $A_n \subseteq U$, so every $a \in A_n$ satisfies $a \leq \sup U$ (as $\sup U$ is an upper bound for U); thus $\sup U$ is an upper bound for A_n , so $\sup A_n \leq \sup U$ (least-upper-bound property of $\sup A_n$). Since this holds for every $n \in \mathbb{N}$, $\sup U$ is an upper bound for S , so $M = \sup S \leq \sup U$. Combining, $\sup U = M = \sup S$.

Case 2: S is not bounded above. Then for every real number K there is some $n \in \mathbb{N}$ with $\sup A_n > K$: if instead every element of A_n were $\leq K$, then K would be an upper bound for A_n , forcing $\sup A_n \leq K$ by the least-upper-bound property, contradicting $\sup A_n > K$; hence some element $x \in A_n \subseteq U$ satisfies $x > K$. Since K was an arbitrary real number, U is also not bounded above, and by the course's convention $\sup U = \infty = \sup S$.

In both cases, $\sup U = \sup S$, so

$$\sup \left(\bigcup_{n=1}^{\infty} A_n \right) = \sup \{ \sup A_n : n \in \mathbb{N} \}.$$

□

Common slip. Quoting the formula without addressing whether $\{\sup A_n\}$ is itself bounded above is an incomplete proof: Case 1's argument needs $\sup S$ to exist before it can be compared with anything. The convention that an unbounded set has supremum ∞ is given as a Remark in Lecture 2, so Case 2 may cite it directly; alternatively, state the formula for the case where $\{\sup A_n\}$ is bounded above and note separately that both sides are unbounded otherwise, as done here.

Unofficial – QRS@NTU – qrs@ntu.org

Question A2 (True/false: nested and non-nested interval intersections)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence of nested open intervals $J_1 \supseteq J_2 \supseteq J_3 \supseteq \cdots$ with $\bigcap_{n=1}^{\infty} J_n$ nonempty but containing only a finite number of elements.
- (b) There is a sequence of closed and bounded (but not necessarily nested) intervals $I_1, I_2, I_3, \dots$ with the property that $\bigcap_{n=1}^N I_n \neq \emptyset$ for all $N \in \mathbb{N}$, but $\bigcap_{n=1}^{\infty} I_n = \emptyset$.

(Tutorial 1, Additional Question A2.)

Solution

Official hint. Answers only: (a) True. (b) False.

Beyond the hints. No justification is given for either part. (a) needs an explicit example verified in full; (b) needs an actual proof (not merely an unsuccessful search for a counterexample) that no such sequence can exist – we give one via the Nested Interval Property, after first reducing to a nested sequence of finite intersections.

Part (a): True

Idea. (*Scratch work – not part of the written answer.*) Take $J_n = (-1/n, 1/n)$. These are nested (each contains the next) and open. The intersection should be $\{0\}$: certainly 0 is in every J_n ; and any nonzero x eventually falls outside J_n once $1/n < |x|$, by the Archimedean Property.

Proof. Let $J_n = (-1/n, 1/n)$ for each $n \in \mathbb{N}$. Since $1/(n+1) < 1/n$, $J_{n+1} \subseteq J_n$ for every $n \in \mathbb{N}$, so $J_1 \supseteq J_2 \supseteq J_3 \supseteq \cdots$ is a nested sequence of open intervals.

We claim $\bigcap_{n=1}^{\infty} J_n = \{0\}$. First, $0 \in J_n$ for every $n \in \mathbb{N}$, since $-1/n < 0 < 1/n$; hence $0 \in \bigcap_{n=1}^{\infty} J_n$, so the intersection is nonempty. Second, let $x \neq 0$ be arbitrary; we show $x \notin \bigcap_{n=1}^{\infty} J_n$. Since $|x| > 0$, the Archimedean Property gives some $n \in \mathbb{N}$ with $1/n < |x|$. Then $x \notin (-1/n, 1/n) = J_n$, because $|x| > 1/n$ places x outside the open interval of radius $1/n$. Since $x \notin J_n$, $x \notin \bigcap_{n=1}^{\infty} J_n$. Therefore no nonzero real number lies in $\bigcap_{n=1}^{\infty} J_n$, so $\bigcap_{n=1}^{\infty} J_n \subseteq \{0\}$; combined with $0 \in \bigcap_{n=1}^{\infty} J_n$, we get $\bigcap_{n=1}^{\infty} J_n = \{0\}$, which is nonempty and contains exactly one (hence finitely many) element.

Statement (a) is TRUE; witnessed by $J_n = (-1/n, 1/n)$, $\bigcap_{n=1}^{\infty} J_n = \{0\}$.

□

Part (b): False

Idea. (*Scratch work – not part of the written answer.*) The stated property (every finite sub-intersection nonempty) is the “finite intersection property”. For closed bounded

intervals, reduce to a nested sequence by intersecting the first N of them: $K_N := I_1 \cap \cdots \cap I_N = [\max(a_1, \dots, a_N), \min(b_1, \dots, b_N)]$ is again a closed bounded interval, nonempty by hypothesis, and $K_1 \supseteq K_2 \supseteq \cdots$ is nested by construction. The Nested Interval Property then forces $\bigcap_N K_N \neq \emptyset$, and $\bigcap_N K_N = \bigcap_n I_n$.

Proof. We prove that no such sequence can exist, which is precisely the statement “(b) is false”. Let $I_n = [a_n, b_n]$, $n \in \mathbb{N}$, be closed and bounded intervals with $\bigcap_{n=1}^N I_n \neq \emptyset$ for every $N \in \mathbb{N}$; we show $\bigcap_{n=1}^\infty I_n \neq \emptyset$.

For each $N \in \mathbb{N}$, define $K_N := \bigcap_{n=1}^N I_n$. We first show $K_N = [\alpha_N, \beta_N]$ where $\alpha_N = \max(a_1, \dots, a_N)$ and $\beta_N = \min(b_1, \dots, b_N)$. Indeed, $x \in K_N$ if and only if $a_n \leq x \leq b_n$ for every $n = 1, \dots, N$, if and only if $\alpha_N \leq x$ and $x \leq \beta_N$, if and only if $x \in [\alpha_N, \beta_N]$. By hypothesis $K_N \neq \emptyset$, so $\alpha_N \leq \beta_N$; hence $K_N = [\alpha_N, \beta_N]$ is a closed and bounded interval.

Next, $K_{N+1} = K_N \cap I_{N+1} \subseteq K_N$ for every $N \in \mathbb{N}$, so $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ is a nested sequence of closed and bounded intervals, each nonempty by hypothesis. By the Nested Interval Property, $\bigcap_{N=1}^\infty K_N \neq \emptyset$.

Finally, $\bigcap_{N=1}^\infty K_N = \bigcap_{N=1}^\infty \bigcap_{n=1}^N I_n = \bigcap_{n=1}^\infty I_n$, since every I_n appears in K_N for all $N \geq n$, so an element lies in every K_N if and only if it lies in every I_n . Therefore $\bigcap_{n=1}^\infty I_n = \bigcap_{N=1}^\infty K_N \neq \emptyset$.

This shows that *every* sequence of closed and bounded intervals with all finite sub-intersections nonempty must also have nonempty full intersection; hence no counterexample to this exists, and statement (b) is false.

Statement (b) is FALSE (via the Nested Interval Property applied to K_N .)

□

Common slip. For part (b), merely failing to find a counterexample is not a proof of falsity; a genuine proof must show that *no* such sequence can exist, as done above via the Nested Interval Property. For part (a), a common slip is choosing *closed* intervals (e.g. $[-1/n, 1/n]$) and citing this as an example of a nested sequence with a finite intersection – that is a different (also true) fact, but the question specifically asks about *open* intervals, where the Nested Interval Property does not directly apply and the intersection must be found by hand.

Question A3 (The set $\mathbb{N} \times \mathbb{N}$ is countable)

Problem

Prove that $\mathbb{N} \times \mathbb{N}$ is countable; i.e., elements of $\mathbb{N} \times \mathbb{N}$ can be listed in a sequence indexed by the natural numbers. (*Tutorial 1, Additional Question A3.*)

Solution

Official hint. “Follow the idea of listing positive rational numbers.”

Beyond the hints. The hint gives no explicit construction. We give the standard diagonal enumeration and a full proof that it is a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ (well-defined, injective, and surjective), which is what “listed in a sequence indexed by the natural numbers” requires.

Method 1: Diagonal enumeration

Idea. (*Scratch work – not part of the written answer.*) Group pairs $(m, n) \in \mathbb{N} \times \mathbb{N}$ by the value $k = m + n - 1$ (the “diagonal index”): diagonal k consists of the k pairs $(1, k), (2, k - 1), \dots, (k, 1)$. Listing diagonal 1, then diagonal 2, then diagonal 3, $\dots$, and within each diagonal listing by increasing m , gives a single sequence covering all of $\mathbb{N} \times \mathbb{N}$ exactly once. The explicit position of (m, n) in this list is $f(m, n) = \frac{(k-1)k}{2} + m$ where $k = m + n - 1$, i.e. (number of pairs in all earlier diagonals) + (position within its own diagonal).

Proof. Define $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ by

$$f(m, n) := \frac{(m + n - 2)(m + n - 1)}{2} + m, \quad m, n \in \mathbb{N}.$$

We show f is a well-defined bijection onto $\mathbb{N}$; this is exactly what is meant by “the elements of $\mathbb{N} \times \mathbb{N}$ can be listed in a sequence indexed by $\mathbb{N}$ ” (the sequence being (m, n) listed in order of $f(m, n) = 1, 2, 3, \dots$).

Step 1: diagonals partition $\mathbb{N} \times \mathbb{N}$. For $(m, n) \in \mathbb{N} \times \mathbb{N}$, set $k = k(m, n) := m + n - 1 \in \mathbb{N}$ (since $m, n \geq 1$, $k \geq 1$). Fixing k , the pairs (m, n) with $m + n - 1 = k$ are exactly $(1, k), (2, k - 1), \dots, (k, 1)$, i.e. m ranges over $1, 2, \dots, k$ with $n = k + 1 - m$ determined by m . Every pair $(m, n) \in \mathbb{N} \times \mathbb{N}$ belongs to exactly one such diagonal (namely $k = m + n - 1$), so the diagonals $D_k := \{(m, n) : m + n - 1 = k\}$, $k = 1, 2, 3, \dots$, partition $\mathbb{N} \times \mathbb{N}$, and D_k has exactly k elements.

Step 2: well-definedness and range on each diagonal. By induction on k , the total number of pairs in $D_1, \dots, D_{k-1}$ is $\sum_{j=1}^{k-1} j = \frac{(k-1)k}{2}$ (the standard triangular-number identity, itself proved by induction: it holds for $k = 1$ vacuously as $0 = 0$, and if $\sum_{j=1}^{k-1} j = \frac{(k-1)k}{2}$ then $\sum_{j=1}^k j = \frac{(k-1)k}{2} + k = \frac{k(k+1)}{2}$). For $(m, n) \in D_k$ (so $k = m + n - 1$ and $1 \leq m \leq k$), $f(m, n) = \frac{(k-1)k}{2} + m$ ranges over $\frac{(k-1)k}{2} + 1, \dots, \frac{(k-1)k}{2} + k = \frac{k(k+1)}{2}$ as m ranges over $1, \dots, k$. In particular $f(m, n) \geq \frac{(k-1)k}{2} + 1 \geq 1$, so $f(m, n) \in \mathbb{N}$; thus f is well-defined as a map into $\mathbb{N}$.

Step 3: injectivity. Suppose $f(m, n) = f(m', n')$. Let $k = m + n - 1$ and $k' = m' + n' - 1$. By Step 2, the values of f on D_k all lie in the integer range $\left(\frac{(k-1)k}{2}, \frac{k(k+1)}{2}\right]$,

and these ranges for distinct diagonals are disjoint (the range for D_k ends exactly where the range for D_{k+1} begins, at $\frac{k(k+1)}{2}$). Since $f(m, n) = f(m', n')$ lies in both the range for D_k and the range for $D_{k'}$, we must have $k = k'$. Within a single diagonal D_k , $f(m, n) = \frac{(k-1)k}{2} + m$ is strictly increasing in m (since the first term is fixed once k is fixed), hence injective in m on D_k ; so $f(m, n) = f(m', n')$ and $k = k'$ together force $m = m'$, and then $n = k + 1 - m = k' + 1 - m' = n'$. Therefore $(m, n) = (m', n')$, so f is injective.

Step 4: surjectivity. Let $p \in \mathbb{N}$ be arbitrary. Since the ranges $\left(\frac{(k-1)k}{2}, \frac{k(k+1)}{2}\right]$ for $k = 1, 2, 3, \dots$ are consecutive integer blocks starting at 1 (the range for D_1 is $(0, 1] = \{1\}$, and the range for D_{k+1} begins immediately after the range for D_k ends, at $\frac{k(k+1)}{2} + 1$) and their union is unbounded (since $\frac{k(k+1)}{2} \rightarrow \infty$ as $k \rightarrow \infty$, by the Archimedean Property), every $p \in \mathbb{N}$ falls into exactly one such range, say the range for D_k . Then $m := p - \frac{(k-1)k}{2}$ satisfies $1 \leq m \leq k$, and setting $n := k + 1 - m \in \mathbb{N}$ gives $(m, n) \in D_k \subseteq \mathbb{N} \times \mathbb{N}$ with $f(m, n) = \frac{(k-1)k}{2} + m = p$. Hence f is surjective.

By Steps 3 and 4, f is a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Listing the pairs (m, n) in increasing order of $f(m, n)$ – equivalently, diagonal by diagonal, and within each diagonal by increasing m – therefore lists every element of $\mathbb{N} \times \mathbb{N}$ exactly once, indexed by $\mathbb{N}$.

$$\boxed{\mathbb{N} \times \mathbb{N} \text{ is countable, via the bijection } f(m, n) = \frac{(m+n-2)(m+n-1)}{2} + m.}$$

□

Common slip. Merely exhibiting the diagonal *picture* (a zig-zag arrow diagram) without an explicit formula or an injectivity/surjectivity argument is not a complete proof that a bijection with $\mathbb{N}$ exists; the course’s own emphasis on “use the definition” extends to definitions of injective/surjective/bijective here. A second slip is forgetting that $\mathbb{N}$ in this course starts at 1, which shifts every index in the formula for f by comparison with a 0-indexed version found in some textbooks.

Method Map

Q.	Core topic	Methods included
1	Sup/inf/max/min of explicit sets	Direct bound + attaining element (max/min cases); ε -Criterion + Archimedean Property (unattained cases).
2	$\max\{\sup A, \sup B\} = \sup(A \cup B)$	Two inequalities from the definition of supremum (exam-safe); ε -Criterion shortcut.
3	$\sup B = \inf A$ for the lower-bound set	Direct verification against the definition of infimum.
4	$\sup(A + B) = \sup A + \sup B$	Upper-bound chase (a); rearrangement (b); least-upper-bound assembly (c).
5	ε -characterisation of $\sup A$	Contrapositive case split via the Archimedean Property.
A1	sup of a countable union	Two inequalities, generalised to countably many sets; case split on boundedness.
A2	Nested vs. non-nested interval intersections	Explicit example (a); Nested Interval Property via finite sub-intersections (b).
A3	Countability of $\mathbb{N} \times \mathbb{N}$	Explicit diagonal-enumeration bijection; injectivity/surjectivity from triangular numbers.

Technique Summary

- Sup/inf proofs are always two-part.** Show the candidate is a bound (upper/lower), then show it is the *least/greatest* such bound – either by exhibiting an attaining element, or via the ε -Criterion for Suprema and Infima together with the Archimedean Property.
- Existence must be checked before use.** $\sup A$ (or $\inf A$) cannot be invoked until A is shown nonempty and bounded above (below); this is exactly the hypothesis of the Axiom of Completeness.
- Two-inequality pattern for combined sets.** Statements of the form “the supremum of a combined set equals a combination of suprema” (Q2, Q4, A1) are proved by showing each side is $\leq$ the other, chasing the definitions of upper bound and supremum in turn.
- The Archimedean Property manufactures indices from gaps.** Given any positive gap ε (or $a \in \mathbb{R}$), the Archimedean Property produces $n \in \mathbb{N}$ with $1/n < \varepsilon$ (or $n > a$); the index is always chosen *after* the gap is fixed.
- The Nested Interval Property turns finite-intersection hypotheses into full-intersection conclusions.** A2(b) shows how to reduce a non-nested family with the finite intersection property to a nested one by taking running intersections.
- Countability proofs need an explicit bijection with verified injectivity and surjectivity,** not just a picture (A3).

Final Answers

Q1(a) $\sup A = \max A = 1$; $\inf A = 0$, not the minimum.

Q1(b) $\sup B = \max B = 1$; $\inf B = \min B = -1$.

Q1(c) $\sup C = \frac{1}{3}$, not the maximum; $\inf C = \min C = \frac{1}{4}$.

Q1(d) $\sup D = 1$, not the maximum; $\inf D = 0$, not the minimum.

Q2 $\max\{\sup A, \sup B\} = \sup(A \cup B)$ – proved.

Q3 $\sup B = \inf A$ – proved (and $\inf A$'s existence established along the way).

Q4 $\sup(A + B) = s + t = \sup A + \sup B$ – proved via (a) upper bound, (b) rearranged bound, (c) least-upper-bound assembly.

Q5 $s = \sup A$ – proved by contrapositive case split.

A1 $\sup(\bigcup_{n=1}^{\infty} A_n) = \sup\{\sup A_n : n \in \mathbb{N}\}$.

A2(a) True – $J_n = (-1/n, 1/n)$ gives $\bigcap_{n=1}^{\infty} J_n = \{0\}$.

A2(b) False – the Nested Interval Property forces nonempty full intersection whenever all finite sub-intersections are nonempty.

A3 $\mathbb{N} \times \mathbb{N}$ is countable via the diagonal bijection $f(m, n) = \frac{(m+n-2)(m+n-1)}{2} + m$.